

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree (S, FE) Examination June 2024 (2019 Scheme)

Course Code: CXT309**Course Name: OBJECT ORIENTED MODELING AND DESIGN**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- | | | |
|----|--|---|
| 1 | Describe the difference between aggregation and composition in object-oriented modelling. | 3 |
| 2 | Explain how object-oriented programming promotes code reusability and maintainability. | 3 |
| 3 | Explain the concept of concurrency in dynamic modeling and its challenges. | 3 |
| 4 | Describe the relationship between object and dynamic models in software development. | 3 |
| 5 | Describe the concept of interface format in dynamic modeling. | 3 |
| 6 | Explain the significance of identifying events in dynamic modeling. | 3 |
| 7 | How the design optimization is performed? | 3 |
| 8 | How the inheritance can be adjusted? | 3 |
| 9 | What are use cases and actors? Explain the use case diagram for an entry subsystem | 3 |
| 10 | Discuss about class diagram and state diagram introduced by Grady Booch's object-oriented analysis and design methodology. | 3 |

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

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|----|--|----|
| 11 | a) Discuss the role of encapsulation, inheritance, and polymorphism in achieving code modularity, reusability, and flexibility in object-oriented programming. | 14 |
| OR | | |
| 12 | a) Differentiate between encapsulation and information hiding in object-oriented programming. | 7 |

- b) Explain the role of polymorphism in supporting dynamic dispatch in object-oriented programming. 7

Module -2

- 13 a) Discuss the significance of dynamic modeling and Nested state diagrams with suitable example. 14

OR

- 14 a) Explain the role of events and states in dynamic modeling. 14

Module -3

- 15 a) Discuss the concept of functional modeling and its role in defining the functional requirements of software systems. 14

OR

- 16 a) Explain the concept of system design and its role in translating software requirements into a high-level system architecture. 14

Module -4

- 17 a) Explain the significance of combining the three models (object, dynamic, and functional) in object design and how it contributes to a comprehensive and cohesive system representation. 14

OR

- 18 a) Explain the concept of documenting design decisions in object design and its importance for maintaining consistency, understanding, and future modifications. 14

Module -5

- 19 a) Discuss the different models involved in Jacobson's methodology and how they contribute to a comprehensive understanding of the software system. 14

OR

- 20 a) Explain the Unified Modeling Language (UML) and its role in object-oriented analysis and design. 14
